European Journal of Operational Research 299 (2022) 1055–1068

Contents lists available at ScienceDirect

European Journal of Operational Research

journal homepage: www.elsevier.com/locate/ejor

Analytics, Computational Intelligence and Information Management

A Mixed Integer Linear Programming Support Vector Machine for Cost-Effective Group Feature Selection: Branch-Cut-and-Price Approach

In Gyu Lee[a], Sang Won Yoon[a,b], Daehan Won[a,*]

[a] Department of Systems Science and Industrial Engineering, State University of New York at Binghamton Binghamton NY 13902, United States
[b] Faculty of Commerce, Waseda University, Tokyo 169–8050, Japan

ARTICLE INFO

Article history:
Received 25 January 2021
Accepted 21 December 2021
Available online 27 December 2021

Keywords:
Machine learning
Feature selection
Support vector machine
Robust optimization
Branch-Cut-and-Price

ABSTRACT

Recently, cost-based feature selection has received significant attention due to its great ability to achieve promising prediction accuracy at a minimum feature acquisition cost. To further improve its predictive and economic performances, this research proposes a cost-effective 1-norm support vector machine with group feature selection as GFS-CESVM$_1$. Its robust counterpart model, GFS-RCESVM$_1$, is also introduced to address the cost uncertainty of features and feature groups because cost variation commonly exists in real-world problems. The proposed models are formulated as Mixed Integer Linear Programming (MILP). To efficiently solve the proposed SVM MILP models, we develop a Branch-Cut-and-Price (BCP) algorithm that considers only a limited number of variables and/or constraints, which thereby leads to rapid convergence to an optimal solution. Various experimental results on benchmark and synthetic datasets demonstrate that GFS-CESVM$_1$ can achieve competitive outcomes by considering not only individual feature evaluation but also group structural information among features. The GFS-RCESVM$_1$ can identify the subset of features that is immune to cost uncertainty and therefore provide feasible and optimal solutions. Furthermore, our BCP algorithm can dominantly outperform the ordinary BB algorithm for finding better objective value and integrality gap within a short period of time.

1. Introduction

Feature selection aims to identify a subset of the most relevant input variables (i.e., features) to enhance model predictability, generalizability, interpretability, and/or lower computational complexity. Due to these advantages, feature selection has emerged as a fundamental data preprocessing step in machine learning (Crone, Lessmann, & Stahlbock, 2006; Piramuthu, 2004; Unler & Murat, 2010). Traditional feature selection schemes focus mainly on improving classification performances while overcoming the curse of dimensionality. However, it ignores a feature acquisition cost (e.g., monetary price, risk, and computational complexity) and therefore lead to unrealistic solutions because of a limited budget in real-world applications. A growing emphasis on cost-effective decision-making in machine learning has motivated researchers to develop cost-based feature selection in order to achieve high accuracy at a minimum cost. That is, a predictive model needs to select only informative yet inexpensive features to correctly predict new or unseen data points. For example, a medical diagnosis in healthcare can be imagined where each medical test's cost is estimated by experts in terms of price, time, implementation difficulty, and side effects (Zhou, Zhou, & Li, 2016). Physicians need to take the most desirable medical tests based on a personal budget to provide for better patients' health and well-being.

Diverse cost-based feature selection models have been developed based on the decision tree, random forest, correlation feature selection, and minimum redundancy and maximum relevance algorithms (Bolón-Canedo, Porto-Díaz, Sánchez-Maroño, & Alonso-Betanzos, 2014; Freitas, Costa-Pereira, & Brazdil, 2007; Zhou et al., 2016). Lately, ℓ_1-Norm Support Vector Machine (ℓ_1-SVM) has been in the spotlight due to its generalizability, flexibility, and global optimality (Lee, Zhang, Yoon, & Won, 2020; Maldonado, Pérez, & Bravo, 2017; Maldonado, Pérez, Weber, & Labbé, 2014). These cost-based feature selection models have shown a great potential to significantly reduce a feature acquisition cost while successfully maintaining high classification accuracy. In particular, Cost-Effective ℓ_1-Norm SVM (CESVM$_1$), which was proposed in our previous study, demonstrated competitive outcomes

* Corresponding author at Department of Systems Science and Industrial Engineering, Binghamton University, 4400 Vestal Parkway E, Binghamton, NY, 13902, United States.

E-mail addresses: ilee13@binghamton.edu (I.G. Lee), yoons@binghamton.edu (S.W. Yoon), dhwon@binghamton.edu (D. Won).

https://doi.org/10.1016/j.ejor.2021.12.030
0377-2217/

in terms of predictive and economic performances (Lee et al., 2020). Also, its robust counterpart model ($RCESVM_1$) addressed feature cost uncertainty because cost variation commonly exists in reality. However, group feature selection was overlooked although real-world data frequently have the structure of grouped features. This research extends $CESVM_1$ and $RCESVM_1$ to further improve the practical aspect by incorporating Group Feature Selection (GFS) into the models as GFS-$CESVM_1$ and GFS-$RCESVM_1$. To efficiently solve the proposed models, a Branch-Cut-and-Price (BCP) is developed.

In real-world applications such as credit scoring, image analysis, and email spam filtering, the structure of grouped features commonly exists (Maldonado et al., 2017; Tang, Adam, & Si, 2018; Wang et al., 2015). Individual feature selection investigates the relative importance of each feature, whereas group feature selection identifies a hidden relationship between feature groups and saves the cost that acquires grouped features for a discounted price. More effective and practical feature selection necessitates considering group structural information among features in addition to evaluating individual features. Many existing models, however, conduct feature selection at either an individual level or a group level although both of them are all important. This research proposes GFS-$CESVM_1$ that simultaneously considers feature selection at both individual and group levels depending on the cost-benefit. If many features in a group are used for prediction, group selection is desired, and otherwise, individual selection is preferred. The proposed model measures significance between individual and group feature selection and flexibly benefits from each advantage. Also, this research formulates the robust model, GFS-$RCESVM_1$, to address the cost uncertainty of individual and grouped features because a small degree of uncertainty can make a nominal optimal solution completely meaningless from a practical viewpoint (Ben-Tal & Nemirovski, 2000). GFS-$RCESVM_1$ identifies optimal and feasible solutions that are more immune to the cost uncertainty and prevent the predictive performance from being significantly deteriorated.

Despite their effectiveness, the proposed SVM models embrace high computational complexity because they are formulated as Mixed Integer Linear Programming (MILP). Integrity in the models can capture the discrete nature of decisions, which cannot be determined by linear programming (LP). The logical representation using integer variables greatly expands the scope of useful optimization problems. However, MILP is computationally expensive with its numerous variables and constraints. To tackle the challenge, we develop a Branch-Cut-and-Price (BCP) algorithm to efficiently solve GFS-$CESVM_1$ and GFS-$RCESVM_1$, which is the main contribution in this research. The BCP algorithm aims to achieve rapid convergence to solution optimality by considering only a limited number of variables and/or constraints in a solution search tree. Specifically, the BCP algorithm starts with restricted variables and constraints. Promising variables and constraints are added to a solution search tree through column and row generation. This process continues until the solution optimality condition is met. There are two main reasons that the proposed BCP algorithm can be greatly beneficial for MILP SVMs: 1) ℓ_1-SVM generally provides sparse solutions by shrinking the weight of irrelevant features towards zero. This feature sparsity requires a small number of variables. 2) By its nature, SVM needs few active constraints associated with support vector samples. This constraint sparsity requires a small number of constraints. These sparsity properties enable a model to solve SVM MILPs in a reasonable time frame by significantly curtailing model complexity and to enhance applicability. Various experiments on benchmark and synthetic datasets are conducted to validate the effectiveness of the proposed SVM models and BCP algorithm. The contributions of this research are summarized as follows:

- GFS-$CESVM_1$ and GFS-$RCESVM_1$ are proposed to further enhance predictive and economic performances with considering group feature selection.
- The BCP algorithm is developed to efficiently address the proposed MILP SVMs with a limited number of variables and constraints on large-scale datasets.
- Various experiments are conducted on benchmark and synthetic datasets to demonstrate the effectiveness of the proposed SVM models and the BCP algorithm.

The remainder of this research is organized as follows: In Section 2, a literature review is presented on the cost-based feature selection models and computationally efficient techniques applied to SVMs. Section 3 describes how to derive the mathematical model of GFS-$CESVM_1$ and GFS-$RCESVM_1$, and Section 4 proposes the BCP algorithm to solve the model. In Section 5, experimental results are analyzed. Lastly, Section 6 concludes the research with future work.

2. Literature review

Recently, the importance of cost-based feature selection, which simultaneously considers misclassification and feature costs, has been emphasized due to its significant effect on predictive and economic performances. To improve practicability, various heuristic and exact solution approaches have been developed. Inexpensive Classification with Expensive Test (ICET) was proposed by adopting a genetic algorithm (GA) in a decision tree framework (Turney, 1994). The GA evolves a population of multiple decision trees based on a fitness function that minimizes misclassification and feature costs. Another GA-based hybrid approach, named distance-based constructive learning algorithm (DistAl), incorporates a neural network (Yang & Honavar, 1998). As GA-based approaches are computationally inefficient as they require the iterative population evolving process, a single decision tree model was proposed where a tree is branched based on the minimal total cost of misclassification and feature acquisition (Ling, Yang, Wang, & Zhang, 2004). However, a single decision tree model may have the issue of overfitting and generalizability. The feature cost-sensitive random forest (FCSRF) was proposed as an ensemble tree learning algorithm, which forms a multitude of decision trees to tackle these issues (Zhou et al., 2016). In FCSRF, the feature cost is included in the tree construction process; therefore, fewer expensive features have higher chances to be selected as a decision node. Additionally, variations of the ordinary feature selection models were proposed from the correlation feature selection (CFS) and the minimum redundancy and maximum relevance (mRMR) (Bolón-Canedo et al., 2014). The feature cost is integrated with the original evaluation function, in which the importance of misclassification and feature costs is controlled by a trade-off parameter λ.

In contrast to the reviewed heuristic methods, SVM has been widely adopted for cost-based feature selection as a mathematical optimization problem due to generalizability and flexibility. The ℓ_1-norm SVM (ℓ_1-SVM) and LP SVM were proposed by incorporating feature acquisition costs into the model formulation (Maldonado et al., 2014). However, in this study, the feature costs were not realistically considered by assuming all the feature costs to be the same as one, although costs are generally different among features in reality. Even the same feature may have a cost variation depending on different situations. The cost uncertainty cannot be ignored as a small degree of uncertainty can make the nominal optimal solution completely meaningless from a practical viewpoint. $RCESVM_1$ was proposed, which considers not only feature cost but also its uncertainty to guarantee the feasibility of any possible solutions (Lee et al., 2020). The SVM models have shown a remarkable ability to significantly reduce expenditure for selecting informative features while maintaining classification accuracy.

The aforementioned models consider the feature acquisition cost at the individual feature level while disregarding the intrinsic group structure of features. However, groups of correlated features commonly exist in real-world applications such as credit scoring, image analysis, and email spam filtering. To tackle this issue, several research works related to group feature selection have been conducted. The group Lasso models that use ℓ_2-regularization were explored (Meier, Van De Geer, & Bühlmann, 2008; Yuan & Lin, 2006). The sparse group Lasso model was also proposed to encourage sparsity at the level of both features and groups (Friedman, Hastie, & Tibshirani, 2010). Similarly, a sparse-group lasso model was introduced, which has sparse effects both on a group and within-group level (Simon, Friedman, Hastie, & Tibshirani, 2013). These models are mainly based on square loss and logistic loss for regression and classification. In addition, a study on group feature selection using a Bayesian method was conducted for linear regression problems (Hernández-Lobato, Hernández-Lobato, & Dupont, 2013). Bayesian group feature selection was also applied to SVMs that explore hinge loss (Du et al., 2016). The SVM with group feature selection was also employed to address multiclass classification (Tang et al., 2018). Another model, Online Multi-label Group Feature Selection (OMGFS), was developed, in which intrinsic group structures of features and streaming feature handling are considered simultaneously (Liu, Lin, Wu, & Wang, 2018). Due to its significance, group feature selection has been recently applied in diverse domains. ℓ_1-SVM and LP SVM were extended by adding group selection of features and tested in the credit scoring domain (Maldonado et al., 2017). Sparse Group Lasso (SGL)-SVM was developed to conduct tumor classification (Huo et al., 2020). Locality-constrained group lasso coding was adopted for microvessel image classification (Chen, Zhou, Kang, & Wen, 2020). While the reviewed group feature selection models were designed to conduct feature selection at a group level, the proposed GFS-CESVM$_1$ can flexibly perform feature selection at both individual and group levels simultaneously depending on the cost-benefit. In this research, we extend CESVM$_1$ to further improve its practical aspect by incorporating group feature selection into the model, GFS-CESVM$_1$, and also propose its robust counterpart, GFS-RCESVM$_1$. However, the proposed SVM models become computationally expensive because they are formulated as MILP by including budget-conscious decision-making for individual and group feature selection along with cost uncertainty. In general, computational complexity in MILP SVMs exponentially increases with the enormous number of features and constraints. Due to the high computational burden of SVM MILPs, most of the related studies in the literature have tested small-scale problems with a few features and constraints.

Only little effort has been made to improve the computational efficiency of MILP SVMs. An algorithm was developed to tighten the bound of the weight vector, which helps to curtail computational complexity by reducing the solution search space (Lee et al., 2020). However, it cannot resolve the essential difficulty caused by numerous features and constraints. Although MILP SVMs are more cumbersome, the majority of studies have focused on improving LP SVMs. A column generation technique was applied to LP SVMs as a boosting method (Bennett, Demiriz, & Shawe-Taylor, 2000; Carrizosa, Martin, & Morales, 2006). The column generation approach was also extended to quadratic programs (QPs) for the mixture of kernels (Bi, Zhang, & Bennett, 2004). The column generation is shown to produce sparse solutions by considering only potentially useful decision variables, thereby significantly reducing the testing time. In addition to column generation, a linear programming chunking algorithm was introduced as a row generation scheme to train the large-scale ℓ_1-SVM (Bradley & Mangasarian, 2000). However, either column or row generation can exploit only one-fold of the sparseness to reduce the number of constraints or variables. A row-column generation method for ℓ_1-SVM was introduced by simultaneously adding optimistic rows and columns and solving the LP problem again (Zhang & Zhou, 2013). An in-depth study is required for MILP SVMs to address large-scale discrete optimization problems. This research develops a BCP algorithm to efficiently solve the proposed MILP SVMs on datasets with large dimensions and sample sizes.

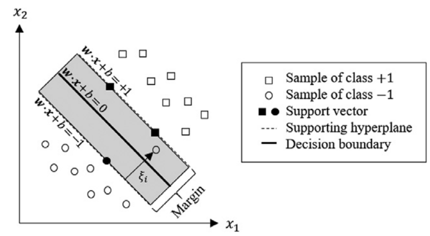


Fig. 1. Illustration of support vector machine.

3. Mathematical models

This section briefly reviews the ℓ_1-SVM model and introduces GFS-CESVM$_1$ and GFS-RCESVM$_1$ that are extended from CESVM$_1$ and RCESVM$_1$.

3.1. ℓ_1-SVM

SVM has been widely used in machine learning, and many variants have been developed. Given empirical data $\{\boldsymbol{x}_i, y_i\}_{i=1}^m$ with training samples $\boldsymbol{x}_i \in \mathbb{R}^l$ (i.e., l-dimensional real spaces) and their corresponding labels $y_i \in \{+1, -1\}$, SVM determines a decision hyperplane; i.e., m and l are the number of samples and features, respectively. The hyperplane, $h(\boldsymbol{x}) = \boldsymbol{w}^\top \cdot \boldsymbol{x} + b$ for $\boldsymbol{w} \in \mathbb{R}^l$, is designed to maximize the margin between different classes as shown in Fig. 1, in which $\boldsymbol{w}$ and b represent the weight vector and bias as decision variables, respectively. An adequate hyperplane (i.e., classifier) can provide high accuracy that correctly categorizes data classes.

One of the most popular forms is ℓ_1-SVM because it is advantageous for feature selection with the following distinctive properties (Zhu, Rosset, Tibshirani, & Hastie, 2004): (1) complexity reduction by using the linear objective function; and (2) sparse solutions with fewer features by shrinking the weight of irrelevant features towards zero. ℓ_1-SVM is formulated as follows:

$$(\ell_1\text{-SVM}) \; \underset{\boldsymbol{w},\xi,b}{\text{minimize}} \;\; \|\boldsymbol{w}\|_1 + C\sum_{i=1}^m \xi_i \tag{3.1}$$

$$\text{s.t.} \;\; y_i(\boldsymbol{w}^\top \cdot \boldsymbol{x}_i + b) \geq 1 - \xi_i, \qquad \forall i, \tag{3.2}$$

$$\xi_i \geq 0, \qquad \forall i. \tag{3.3}$$

Slack variables, ξ_i for $i = \{1, \ldots, m\}$, penalize incorrectly classified data points, and a regularization parameter C controls the training error against margin maximization. While a large value of C forces the model to focus on minimizing the misclassification, a small value of C focuses more on the 1-norm regularization of the weight vector. Two non-negative vectors $\boldsymbol{w}^+$ and $\boldsymbol{w}^-$ are imposed to linearize the weight vector $\boldsymbol{w}$. Then, ℓ_1-SVM can be rewritten without loss of generality as follows:

$$(\ell_1\text{-SVM}) \; \underset{\boldsymbol{w}^+,\boldsymbol{w}^-,\xi,b}{\text{minimize}} \;\; \sum_{j=1}^l (w_j^+ + w_j^-) + C\sum_{i=1}^m \xi_i \tag{3.4}$$

$$\text{s.t.} \quad y_i\left(\sum_{j=1}^l ((w_j^+ - w_j^-) \cdot x_{ij}) + b\right) \geq 1 - \xi_i, \qquad \forall i, \tag{3.5}$$

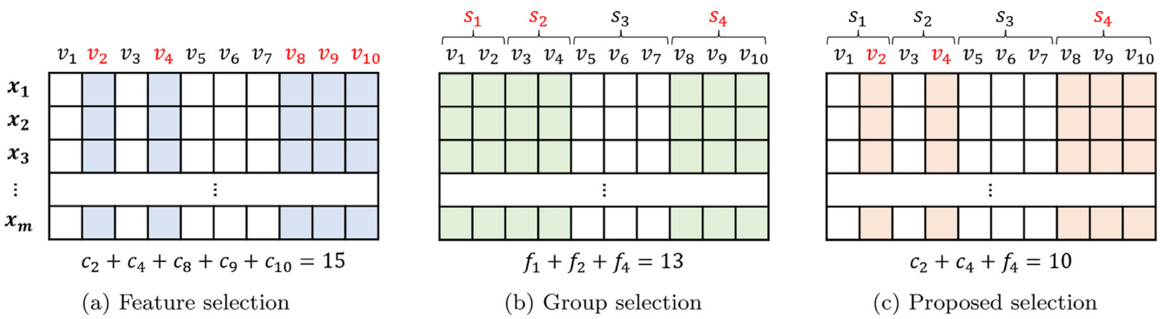


Fig. 2. Comparison of feature selection, group selection, and proposed selection methods, i.e., the selected features and groups are denoted in red, and the corresponding columns are highlighted. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$w_j^+, w_j^- \geq 0, \qquad \forall j, \tag{3.6}$$

$$\xi_i \geq 0, \qquad \forall i. \tag{3.7}$$

3.2. Cost-Based feature selection model

In real-world applications such as credit scoring, image analysis, and email spam filtering, groups of correlated features commonly exist. Individual feature selection investigates the relative importance of each feature, whereas group feature selection identifies a hidden relationship between features and saves the cost that acquires grouped features for a discounted price. It is noted that group structural information among features is known a priori. Therefore, more effective models need to consider not only individual feature evaluation but also the group structure. Many existing models, however, have been designed to conduct feature selection at either an individual level or a group level. Given the cost of individual features and groups, this research proposes GFS-CESVM$_1$ that simultaneously performs feature selection at both individual and group levels depending on the cost-benefit. That is, the proposed model selects individual features in a group if they are cheaper than selecting the group; otherwise, the model selects the grouped features for a group cost. Fig. 2 illustrates the individual feature selection, group selection, and proposed selection methods, in which v_j and s_k denote each feature and group, and c_j and f_k represent their corresponding costs, respectively. Suppose that the optimal set of features for classification is $\{v_2, v_4, v_8, v_9, v_{10}\}$. In this example, the individual feature selection identifies the five features with the cost of 15; the group selection identifies three groups $\{s_1, s_2, s_4\}$ with the cost of 13; and the proposed selection identifies two features and one group $\{v_2, v_5, s_4\}$ with the cost of 10. Therefore, GFS-CESVM$_1$ has a great potential to successfully achieve high accuracy at a minimum cost when a budget is limited. The model formulation is as follows:

$$(\text{GFS-CESVM}_1) \ \underset{\boldsymbol{w}^+, \boldsymbol{w}^-, \boldsymbol{v}, \boldsymbol{s}, \boldsymbol{\xi}, b}{\text{minimize}} \qquad (3.4)$$

$$\text{s.t.} \sum_{k=1}^{n} \left(\sum_{j \in G_k} c_j (v_j - s_k v_j) + f_k s_k \right) \leq B, \tag{3.8}$$

$$\sum_{j \in G_k} c_j v_j \leq W_k s_k + f_k, \qquad \forall k, \tag{3.9}$$

$$w_j^+ + w_j^- \leq M_j v_j, \qquad \forall j, \tag{3.10}$$

$$s_k \in \{0, 1\}, \qquad \forall k, \tag{3.11}$$

$$v_j \in \{0, 1\}, \qquad \forall j, \tag{3.12}$$

(3.5)–(3.7).

Eq. (3.8) is a budget constraint to ensure that the subset of selected features does not exceed a predefined budget of B. Binary decision variables, v_j and s_k, denote the feature selection of the jth individual feature and the kth feature group, respectively, and G_k represents the set of features in the kth group; e.g., $G_1 = \{j: 1, 2, 3\}$, $G_2 = \{j: 4\}$, $G_3 = \{j: 5, 6\}$, and $G_4 = \{j: 7, 8, 9, 10\}$. The term of $\sum_{j \in G_k} c_j (v_j - s_k v_j)$ calculates the total cost of individually selected features in the kth group, while the term of $f_k s_k$ calculates the group cost. The term of $(v_j - s_k v_j)$ prevents the cost from being double-counted; i.e., when the kth group is selected ($s_k = 1$), the cost of individually selected features in the group ($v_j = 1$ for $j \in G_k$) is not counted, and otherwise, the cost of individually selected features is only counted. Eq. (3.9) activates or deactivates individual feature selection depending on the benefit of paying the group cost. If only a few individual features in a group are used, the group cost may not be preferred despite a discounted group price. The input parameter W_k needs to be sufficiently large at least the total cost of all features in the kth group. Eq. (3.10) activates or deactivates weight vectors, w_j^+ and w_j^-, based on selected features signified by v_j. Also, M_j needs to be sufficiently large enough to guarantee solution optimality and feasibility. The weight vectors (w_j^+ and w_j^-) are explored within the bound of M_j. The current formulation includes the product of two binary variables, $s_k v_j$, in (3.8), and r_{kj} is introduced to linearize the nonlinear term. Without loss of generality, GFS-CESVM$_1$ can be transformed as follows:

$$(\text{GFS-CESVM}_1) \ \underset{\boldsymbol{w}^+, \boldsymbol{w}^-, \boldsymbol{v}, \boldsymbol{\xi}, \boldsymbol{s}, \boldsymbol{r}, b}{\text{minimize}} \qquad (3.4)$$

$$\text{s.t.} \quad \sum_{k=1}^{n} \left(\sum_{j \in G_k} c_j (v_j - r_{kj}) + f_k s_k \right) \leq B, \tag{3.13}$$

$$r_{kj} \leq v_j, \qquad \forall k, \forall j \in G_k, \tag{3.14}$$

$$r_{kj} \leq s_k, \qquad \forall k, \forall j \in G_k, \tag{3.15}$$

$$r_{kj} \geq v_j + s_k - 1, \qquad \forall k, \forall j \in G_k, \tag{3.16}$$

$$0 \leq r_{kj} \leq 1, \qquad \forall k, \forall j \in G_k, \tag{3.17}$$

(3.5)-(3.7), (3.9)-(3.12).

It is common that the feature cost has variation because the quantification of the specific feature price is tricky in reality. Therefore, we take into account cost uncertainty in GFS-CESVM$_1$ to derive the robust counterpart model GFS-RCESVM$_1$.

3.3. Robust counterpart model

The proposed GFS-RCESVM$_1$ aims to find the best worst-case solutions to guarantee feasibility in any possible solutions. This paradigm can, however, lead to over-conservatism by abandoning a considerable degree of optimality. This research employs a robust optimization technique that offers full control over the degree of conservatism against robustness, which was proposed in (Bertsimas & Sim, 2004). The individual feature and group costs are assumed to be bounded; i.e., $\tilde{c}_j = [c_j - \hat{c}_j, c_j + \hat{c}_j]$ for $j \in U$ and $\tilde{f}_k = [f_k - \hat{f}_k, f_k + \hat{f}_k]$ for $k \in D$ where $U = \{j\colon\ \hat{c}_j > 0\}$ and $D = \{k\colon\ \hat{f}_k > 0\}$, respectively. The uncertainty level (i.e., the number of uncertain variables) is controlled by the parameter Γ for individual features in the interval $[0, |U|]$ and by the parameter Ω for feature groups in the interval $[0, |D|]$. A solution is more conservative as Γ is closer to $|U|$ and Ω is closer to $|D|$, while the problem becomes identical to the nominal problem when $\Gamma = 0$ and $\Omega = 0$. GFS-RCESVM$_1$ is formulated by adding the individual feature cost uncertainty term, $\beta(\boldsymbol{v}, \boldsymbol{r}, \Gamma) = \max_{\{S: S\subseteq U, |S|\le\Gamma\}} \sum_{k=1}^{n} \sum_{j\in S\cap G_k} \hat{c}_k(v_j - r_{kj})$, and the group cost uncertainty term, $\beta(\boldsymbol{g}, \Omega) = \max_{\{H: H\subseteq D, |H|\le\Omega\}} \sum_{k\in H} \hat{f}_k s_k$, to the budget constraint in (3.13).

$$\underset{\boldsymbol{w}^+, \boldsymbol{w}^-, \boldsymbol{v}, \boldsymbol{g}, \boldsymbol{r}, \boldsymbol{\xi}, b}{\text{minimize}} \quad (3.4)$$

$$\text{s.t.} \quad \sum_{k=1}^{n} \sum_{j\in G_k} c_j(v_j - r_{kj}) + \max_{\{S: S\subseteq U, |S|\le\Gamma\}} \sum_{k=1}^{n} \sum_{j\in S\cap G_k} \hat{c}_k(v_j - r_{kj})$$
$$+ \sum_{k=1}^{n} f_k s_k + \max_{\{H: H\subseteq D, |H|\le\Omega\}} \sum_{k\in H} \hat{f}_k s_k \le B, \tag{3.18}$$
$$\text{(3.5)-(3.7), (3.9)-(3.12), (3.14)-(3.17).}$$

$\beta(\boldsymbol{v}, \boldsymbol{r}, \Gamma)$ and $\beta(\boldsymbol{g}, \Omega)$ protect solution feasibility against uncertainty; however, they make the model complex as the model becomes nonlinear programming problems. To linearize $\beta(\boldsymbol{v}, \boldsymbol{r}, \Gamma)$ and $\beta(\boldsymbol{g}, \Omega)$, the following proposition is considered. We refer interested readers to (Bertsimas & Sim, 2004).

Proposition 1. *Given vectors $\boldsymbol{v}^*$ and $\boldsymbol{s}^*$, the functions $\beta(\boldsymbol{v}^*, \boldsymbol{r}^*, \Gamma)$ and $\beta(\boldsymbol{s}^*, \Omega)$ are equivalent to the following linear optimization problems:*

$$\beta(\boldsymbol{v}^*, \boldsymbol{r}^*, \Gamma) = \underset{\boldsymbol{z}}{\text{maximize}} \ \sum_{k=1}^{n} \sum_{j\in U\cap G_k} \hat{c}_k(v_j^* - r_{kj}^*) z_j \tag{3.19}$$

$$\text{s.t.} \quad \sum_{j\in U} z_j \le \Gamma, \tag{3.20}$$

$$0 \le z_j \le 1, \quad \forall j \in U. \tag{3.21}$$

$$\beta(\boldsymbol{s}^*, \Omega) = \underset{\boldsymbol{t}}{\text{maximize}} \ \sum_{k\in D} \hat{f}_k s_k^* t_k \tag{3.22}$$

$$\text{s.t.} \quad \sum_{k\in D} t_k \le \Omega, \tag{3.23}$$

$$0 \le t_k \le 1, \quad \forall k \in D. \tag{3.24}$$

Proof. *The optimal solution value of the objective function in (3.19) comprises Γ variables of z_j at one. It is equivalent to selecting the subset of features, $\{S\colon S \subseteq U, |S| \le \Gamma\}$, with the corresponding cost function $\sum_{k=1}^{n} \sum_{j\in S\cap G_k} \hat{c}_k(v_j^* - r_{kj}^*)$. Similarly, the optimal solution value of the objective function in (3.22) comprises Γ variables of t_k at one. It is equivalent to selecting the subset of groups, $\{H : H \subseteq D, |H| \le \Omega\}$, with the corresponding cost function $\sum_{k\in H} \hat{f}_k s_k^*$.* □

Theorem 1. *The nonlinear formulation of* GFS-RCESVM$_1$ *can be converted into the following MILP formulation:*

$$\text{(GFS-RCESVM}_1\text{)} \quad \underset{\boldsymbol{w}^+, \boldsymbol{w}^-, \boldsymbol{v}, \boldsymbol{s}, \boldsymbol{r}, \boldsymbol{p}, \boldsymbol{o}, q, h, \boldsymbol{\xi}, b}{\text{minimize}} \quad (3.4)$$

$$\text{s.t.} \quad \sum_{k=1}^{n} \sum_{j\in G_k} c_j(v_j - r_{kj}) + \sum_{j\in U} p_j + q\Gamma$$
$$+ \sum_{k=1}^{n} f_k s_k + \sum_{k\in D} o_k + h\Omega \le B, \tag{3.25}$$

$$q + p_j \ge \hat{c}_j(v_j - r_{kj}), \quad \forall k \in D, \forall j \in G_k \tag{3.26}$$

$$h + o_k \ge \hat{f}_k s_k, \quad \forall k \in D, \tag{3.27}$$

$$p_j \ge 0, \quad \forall j \in U, \tag{3.28}$$

$$o_k \ge 0, \quad \forall k \in D, \tag{3.29}$$

$$q, h \ge 0, \tag{3.30}$$
$$\text{(3.5)-(3.7), (3.9)-(3.12), (3.14)-(3.17).}$$

Proof. *Consider the dual form of the problem in (3.19)–(3.21) and the problem in (3.22)–(3.24):*

$$\underset{\boldsymbol{p}, q}{\text{minimize}} \ \sum_{j\in U} p_j + q\Gamma \tag{3.31}$$

$$\text{s.t.} \quad q + p_j \ge \hat{c}_j(v_j^* - r_{kj}^*), \quad \forall k \in D, \forall j \in G_k \tag{3.32}$$

$$p_j \ge 0, \quad \forall j \in U, \tag{3.33}$$

$$q \ge 0. \tag{3.34}$$

$$\underset{\boldsymbol{o}, h}{\text{minimize}} \ \sum_{k\in D} o_k + h\Omega \tag{3.35}$$

$$\text{s.t.} \quad h + o_k \ge \hat{f}_k s_k^*, \quad \forall k \in D, \tag{3.36}$$

$$o_k \ge 0, \quad \forall k \in D, \tag{3.37}$$

$$h \ge 0. \tag{3.38}$$

By strong duality, the problem in (3.19)–(3.21) is feasible and bounded for $\Gamma \in [0, |U|]$, and the dual problem in (3.31)–(3.34) is also feasible and bounded; therefore, their objective values coincide. Also, the problem in (3.22)–(3.24) is feasible and bounded for $\Omega \in [0, |D|]$, and the dual problem in (3.35)–(3.38) is also feasible and bounded; therefore, their objective values coincide. Consequently, $\beta(\boldsymbol{v}^*, \boldsymbol{r}^*, \Gamma)$ and $\beta(\boldsymbol{s}^*, \Omega)$ can be linearized and substituted in GFS-RCESVM$_1$. □

4. Solution approach: Branch-Cut-and-Price algorithm

Although GFS-RCESVM$_1$ can provide a cost-effective solution that is optimal and feasible from any cost variations, this MILP SVM model could be computationally burdensome with its enormous number of variables and constraints. This section proposes a BCP algorithm to efficiently solve GFS-RCESVM$_1$, which thereby expedites problem-solving. The BCP algorithm is an LP-based BB technique that finds an optimal solution with only a limited number of variables and constraints. There are two main reasons that the proposed BCP algorithm can be greatly beneficial for MILP SVMs: 1) ℓ_1-SVM generally provides sparse solutions by shrinking the weight of irrelevant features towards zero. This feature sparsity requires only a small number of variables. 2) By its nature, the SVM needs few active constraints associated with support vector samples. Similarly, this constraint sparsity requires only a small number of constraints. These sparsity attributes curtail a solution search space, which results in rapid convergence to solution optimality.

Let $\mathcal{I} = \{i: 1, \ldots, m\}$ and $\mathcal{J} = \{j: 1, \ldots, l\}$ be the set of all training samples and features, respectively, and I and J be their subsets that are relatively smaller, i.e., $I \subset \mathcal{I}$ and $J \subset \mathcal{J}$ where $|I| \ll |\mathcal{I}|$ and $|J| \ll |\mathcal{J}|$. Instead of solving GFS-RCESVM$_1$ based on $\mathcal{I}$ and $\mathcal{J}$ as a *master problem* (MP), the proposed BCP algorithm addresses a *restricted master problem* (RMP) using I and J. If the current optimal solution of RMP does not satisfy optimality to MP, a subproblem, *column pricing problem* (CPP), is solved to identify a new feature that could improve an objective value and add it to J. As the new feature is included, the *row pricing problem* (RPP) identifies new samples associated with support vectors that determine an optimal separating hyperplane and added them to I. Variables and constraints, which are corresponding to the newly found feature and samples, are incorporated into RMP. If no promising samples and/or features are discovered, nodes are branched to explore the solution search tree further or fathomed. This process is repeated until the solution optimality to MP is satisfied. It is worth noted that the proposed BCP can be also simply applied to GFS-CESVM$_1$ by manipulating MP and RMP correspondingly.

4.1. Initial subset of feature variables and constraints

The initial feature subset's quality is crucial to rapidly find an optimal solution because irrelevant features may lead to more extended exploration in a solution search tree. We employ the one-way Analysis of Variance (ANOVA) F-test feature selection model to identify a proper subset of initial features. The features are ranked by sorting the F-value in ascending order, and high-ranking features are included in the initial feature subset until the total cost does not exceed a predefined budget B. The ANOVA has been used for feature selection with several advantages (Ding, Feng, Chen, & Lin, 2014; Elssied, Ibrahim, & Osman, 2014; Kumar, Rath, Swain, & Rath, 2015): (1) it is intuitive to analyze the interaction between each feature and the target vector; (2) it can be used when the number of samples is different across features; (3) it can be generalized to more than two groups without increasing the Type I error; and (4) it can be applied when negative values are presented in input data.

In SVM, the vast majority of constraints comprise margin constraints, $y_i(\boldsymbol{w}^\top \cdot \boldsymbol{x}_i + b) \geq 1 - \xi_i$ for $i \in \{1, \ldots, m\}$, which indicate whether samples are on the correct side of the margin or not. An optimal hyperplane is affected by samples lying on or within the margin. Such samples are highly dependent on newly added features from column generation as the BCP algorithm proceeds. Therefore, the effect of the initial subset of samples is insignificant, and a random sampling method is adopted to decide the initial samples due to its simplicity and lack of bias.

4.2. Column generation

Column generation is a well-known mathematical programming technique that solves a series of reduced problems with restricted columns, in which a promising feature variable is added to RMP as required. To identify a new feature that needs to be included for solution optimality, CPP is developed based on the restricted dual of ℓ_1-SVM with the subsets I and J.

$$\text{(Restricted Dual of } \ell_1\text{-SVM)} \ \underset{\boldsymbol{\pi}}{\text{maximize}} \sum_{i \in I} \pi_i \tag{4.1}$$

$$\text{s.t.} \quad -1 \leq \sum_{i=1}^{m} \pi_i y_i x_{ij} \leq 1, \quad \forall j \in J, \tag{4.2}$$

$$\sum_{i \in I} \pi_i y_i = 0, \tag{4.3}$$

$$0 \leq \pi_i \leq C, \qquad \forall i \in I. \tag{4.4}$$

Let $\hat{\boldsymbol{\pi}}$ be the current solution of the restricted dual of ℓ_1-SVM. If $\hat{\boldsymbol{\pi}}$ is not optimal to the dual problem with all constraints and feature variables from $\mathcal{I}$ and $\mathcal{J}$, dual variables that violate the constraints indicate promising features. A feature with the most negative reduced cost is added to J, and the associated variables are incorporated into RMP with the prospect of improving the objective value at the next node. The reduced cost of each feature variable, $\rho(j)$, is computed as follows:

$$\rho(j) = 1 - \left| \sum_{i \in I} \hat{\pi}_i y_i x_{ij} \right|, \quad \forall j \notin J. \tag{4.5}$$

4.3. Row generation

As a new feature is added, row generation needs to discover all samples that could contribute to the construction of a maximal separating hyperplane and add them to I. To identify the promising samples, RPP is developed based on the restricted primal of ℓ_1-SVM with the subsets I and J.

$$\text{(Restricted Primal of } \ell_1\text{-SVM)} \ \underset{\boldsymbol{w}^+, \boldsymbol{w}^-, \boldsymbol{\xi}, b}{\text{minimize}} \ \sum_{j \in J} (w_j^+ + w_j^-) + C \sum_{i \in I} \xi_i \tag{4.6}$$

$$\text{s.t.} \quad y_i \Big(\sum_{j \in J} ((w_j^+ - w_j^-)^\top \cdot x_{ij}) + b \Big) \geq 1 - \xi_i, \qquad \forall i \in I, \tag{4.7}$$

$$w_j^+, w_j^- \geq 0, \qquad \forall j \in J, \tag{4.8}$$

$$\xi_i \geq 0, \qquad \forall i \in I. \tag{4.9}$$

Let $(\hat{\boldsymbol{w}}, \hat{b})$ be the current solution of the restricted primal of ℓ_1-SVM. Samples lying on or within the margin, which satisfy $0 \leq y_i(\hat{\boldsymbol{w}}^\top \cdot \boldsymbol{x}_i + \hat{b}) \leq 1$, influence the hyperplane quality. Such samples (i.e., $\tau(i) \geq 0$) are added to I, and the corresponding margin constraints are incorporated into RMP, in which $\tau(i)$ is computed as follows:

$$\tau(i) = 1 - y_i \sum_{j \in J} (\hat{\boldsymbol{w}}^\top \cdot \boldsymbol{x}_i + \hat{b}), \quad \forall i \notin I. \tag{4.10}$$

Proposition 2. *The ℓ_1-SVM can be used to identify promising features and samples for* GFS-RCESVM$_1$.

Proof. GFS-RCESVM$_1$ is the MILP model derived from the LP model of ℓ_1-SVM by adding the budget constraint with cost uncertainty protection terms while retaining the SVM principle that constructs a maximal separating hyperplane. That is, the only difference between the two models is the presence of the budget constraint. Any integer feasible point is always a lower bound on the optimal LP objective value, and therefore, important variables and constraints are all considered in GFS-RCESVM$_1$ at the optimality of ℓ_1-SVM. □

Proposition 3. *The optimality of the proposed BCP algorithm can be verified by ensuring that ℓ_1-SVM with I and J satisfies optimality to that with $\mathcal{I}$ and $\mathcal{J}$.*

Proof. Let $(\boldsymbol{w}^*, b^*)$ and $\boldsymbol{\pi}^*$ be the optimal solutions to the primal and dual problems of ℓ_1-SVM, respectively, with I and J. By strong duality, if primal and dual solutions are feasible and bounded, their optimal objective function values coincide. Therefore, if $\boldsymbol{\pi}^*$ is optimal to the dual of ℓ_1-SVM with $\mathcal{I}$ and $\mathcal{J}$, the constraints in the primal of ℓ_1-SVM cannot be violated; that is, the following constraints will hold, and the solution optimality and feasibility are verified.

$$-1 \leq \sum_{i=1}^{m} \pi_i^* y_i x_{ij} \leq 1 \quad \forall j \in \mathcal{J}. \tag{4.11}$$

□

4.4. Branching and node selection rules

Various branching techniques have been developed, such as most-infeasible branching, pseudo-cost branching, strong branching, reliability branching, etc. In this research, we employ reliability branching that is a hybrid method that uses pseudo-cost branching and strong branching, which is introduced by Achterberg, Koch, & Martin (2005). Strong branching can efficiently reduce the size of the solution search tree; however, it is computationally expensive. Pseudo-cost branching uses statistics from the current tree, but it has insufficient information for statistics at the beginning of the tree. In reliability branching, the strong branching rule is applied only to the tree's initial nodes to compensate for the lack of statistical information. Then, the pseudo-cost branching rule uses the information to estimate the improvement of lower bound per unit of change in a branching variable in subsequent nodes. By addressing the drawback of strong branching and pseudo-cost branching, reliability branching has been one of the most common techniques. We refer interested readers to (Achterberg et al., 2005; Belotti, Lee, Liberti, Margot, & Wächter, 2009; Gamrath et al., 2016). Also, the node selection strategy used in this research is a best-first search that expands fewer nodes than other admissible algorithms but typically requires larger space. The detailed procedure of the proposed BCP is presented in Algorithm 1.

Algorithm 1: A BCP algorithm for GFS-RCESVM$_1$.

```
1  Input: Initial subset of samples and features, I and J;
2  Output: Optimal classifier that consists of w* and b*;
3  Solve the LP relaxation of RMP with I and J and find the
   current dual optimal solution π̂
4  Check solution optimality to MP by
   −1 ≤ Σ_{i∈I} π̂_i y_i x_ij ≤ 1 ∀j ∈ J;
5  if MP optimality is not satisfied then
6  |  Solve CPP and add a feature with the most negative
   |  reduced cost to J,
7  |  ψ = min{j: ρ(j) < 0, ∀j ∉ J} where
   |  ρ(j) = 1 − |Σ_{i∈I} π̂_i y_i x_ij|, ∀j ∉ J;
8  |  Solve RPP and add all samples that are potentially
   |  associated with support vectors to I,
9  |  Θ = {i: τ(i) ≥ 0, ∀i ∉ I} where
   |  τ(i) = 1 − y_i Σ_{j∈J}(ŵ^⊤ · x_i + b̂), ∀i ∉ I;
10 |  if New feature and/or constraints is found then
11 |  |  Add a new feature variable, v_ψ, to RMP;
12 |  |  Add new constraints,
   |  |  y_i(Σ_{j=1}^{l}((w_j^+ − w_j^−)^⊤ · x_ij) + b) ≥ 1 − ξ_i, ∀i ∈ Θ, to
   |  |  RMP;
13 |  else if Solution is not integer or better than the incumbent
   |  solution then
14 |  |  Branch the search tree based on branching and node
   |  |  selection rules in Section 4.4;
15 |  else Fathom nodes;
16 |  Go to the line 3;
17 else Terminate the algorithm.
```

5. Experimental results and analyses

To validate the effectiveness of the proposed MILP SVMs and BCP algorithm, this sections test seven benchmark datasets from the UCI repository (Asuncion & Newman, 2007) and three synthetic datasets. The proposed models and algorithm are implemented using PySCIPOpt that is a Python interface to the SCIP optimization software. The detailed conditions for computation are Python 3.7 on an Intel Core i7 CPU @3.40 GHz with 24.0 GB RAM running Windows 10.

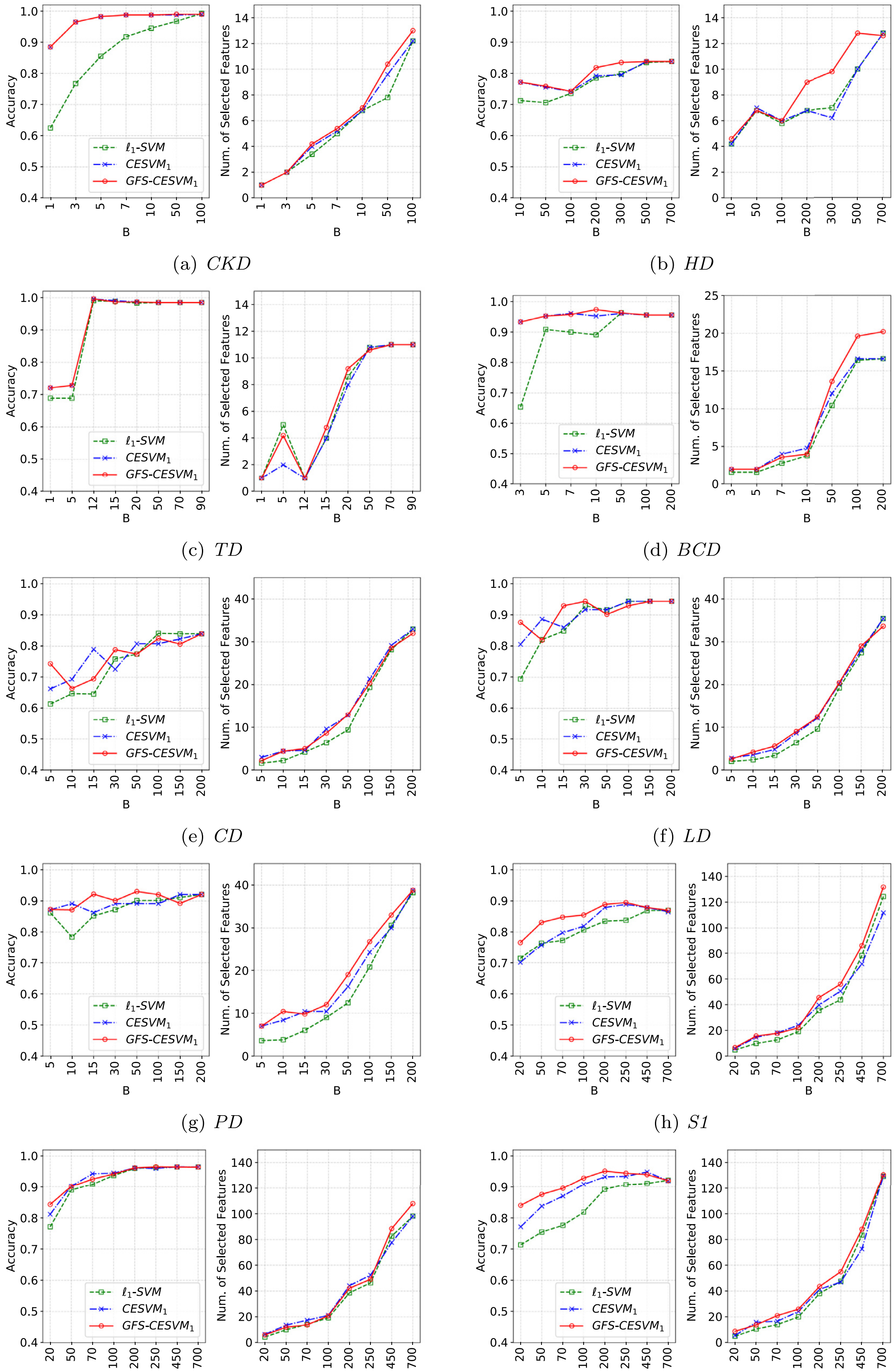
5.1. Dataset description and experimental setup

Table 1 summarizes the experimental datasets that vary in the feature dimension and sample sizes are significantly different. This research focuses mainly on medical diagnosis datasets, and the related benchmark datasets are mostly characterized by 1) low dimensions and large samples such as *CKD*, *HD*, *TD*, and *BCD* and 2) high dimensions and small samples such as *CD*, *LD*, and *PD*. The significant imbalance between the feature dimension and sample sizes encouraged us to conduct more rigorous tests with synthetic datasets, *S1*, *S2*, and *S3*, which are generated with different settings. Specifically, *S1*, *S2*, and *S3* include 10, 20, 30% of informative features to classification, respectively. Also, there is a limitation on the feature cost in some datasets. In *CKD*, *HD*, and *TD*, the cost is estimated by experts (Turney, 2000); however, the cost information is not given for the rest. A random cost between 1 and 10 is generated for each feature using a uniform distribution that is a typical technique used in other related works to tackle this issue (Bolón-Canedo et al., 2014; Lee et al., 2020; Zhou et al., 2016); i.e., $c_j = \mathrm{U}(1, 10)$ for $j = \{1, \ldots, l\}$. In addition, we assume that a group can comprise the maximum of three features for the low-dimension datasets such as *CKD*, *HD*, *TD*, and *BCD*, while a group can comprise the maximum of 10% of features for the other datasets. In general, a group cost is lower than a total cost of individual features in the group due to a discounted price; i.e., $f_k < \sum_{j\in G_k} c_j$. We diversify experiments by varying a group cost ratio to a total cost of individual features across the datasets from 60% to 90%; e.g., $f_k = 0.9 \times \sum_{j\in G_k} c_j$. Moreover, individual features and groups are assumed to have 50% of the cost variation; i.e., $\tilde{c}_j = [c_j - \hat{c}_j, c_j + \hat{c}_j]$ where $\hat{c}_j = 0.5 \times c_j$ for $j = \{1, \ldots, l\}$ and $\tilde{f}_k = [f_k - \hat{f}_k, f_k + \hat{f}_k]$ where $\hat{f}_k = 0.5 \times f_k$ for $k = \{1, \ldots, n\}$.

To conduct the experiments, there are several parameters to be predefined. M_j for $j \in \{1, \ldots, l\}$ and W_k for $k = \{1, \ldots, n\}$ are set to 1000 because it is large enough to guarantee solution optimality and feasibility on our experimental datasets. For the initial subset of constraints, 20% of samples are randomly selected, and the associated margin constraints are incorporated into the model. As in (Lee et al., 2020), we define the regularization parameter C as one, the optimality gap as zero, and the solving time limit as 600 s. Lastly, the experiments are conducted based on 5-fold cross-validation that splits a dataset into five-folds, each of which is used as a testing set, and the other four folds are used as training sets. The cross-validation method has been widely used to appraise the generalization ability of a predictive model and to avoid overfitting. Also, it helps the model properly tune the model classifier parameters, such as w_j and b.

5.2. Experimental results of feature selection

This section investigates the significance of considering the feature acquisition cost in feature selection at individual and group levels simultaneously by comparing GFS-CESVM$_1$ to ℓ_1-SVM and CESVM$_1$. The feature cost is not embodied in ℓ_1-SVM, and therefore, features with the highest weight vector are selected under a predefined budget in a heuristic manner. The CESVM$_1$ only considers the individual feature cost without considering the structure of grouped features. Conversely, the proposed GFS-CESVM$_1$ conducts feature selection at the individual and group levels based on the cost-benefit. Table 2 summarizes the model performances in the classification accuracy, Matthew correlation coefficient (MCC), number of selected features, and total cost. The accuracy and MCC gauge how correctly a model classifies test data points. In general,



(a) *CKD* (b) *HD*

(c) *TD* (d) *BCD*

(e) *CD* (f) *LD*

(g) *PD* (h) *S1*

(i) *S2* (j) *S3*

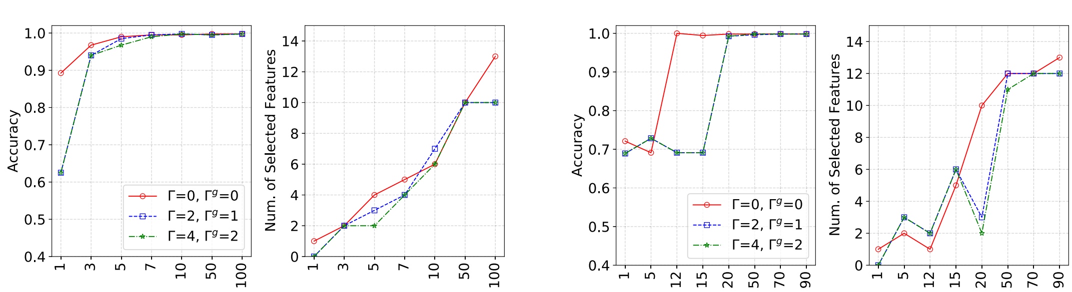
Fig. 3. Performance behaviors of cost-based feature selection models under different budgets.

Table 1
Summary of experimental datasets.

Datasets	Feature size	Sample size	Feature cost	Group size	Group cost ratio
Chronic Kidney Disease (CKD)	24	400	Given	⌈3⌉	60%
Heart Disease (HD)	13	303	Given	⌈3⌉	70%
Thyroid Disease (TD)	20	534	Given	⌈3⌉	80%
Breast Cancer Disease (BCD)	30	569	U(1,10)	⌈3⌉	90%
Colon Disease (CD)	2000	62	U(1,10)	⌈20⌉	70%
Leukemia Disease (LD)	7070	72	U(1,10)	⌈71⌉	80%
Prostate Disease (PD)	5966	102	U(1,10)	⌈60⌉	90%
Synthetic 1 (S1)	2000	1000	U(1,10)	⌈20⌉	70%
Synthetic 2 (S2)	1000	2000	U(1,10)	⌈10⌉	80%
Synthetic 3 (S3)	1500	1500	U(1,10)	⌈15⌉	90%

Table 2
Performance comparison in terms of the accuracy, MCC, number of selected features, and total cost; i.e., the mean value of five folds is measured with standard deviation denoted as σ.

Dataset	Model	Accuracy (σ)	MCC (σ)	# features (σ)	Total cost (σ)
CKD	ℓ_1-SVM	0.97 (0.03)	0.93 (0.06)	7.80 (1.48)	47.78 (3.40)
	CESVM$_1$	0.98 (0.01)	0.96 (0.03)	4.00 (0.00)	4.65 (0.00)
	GFS-CESVM$_1$	0.98 (0.01)	0.96 (0.03)	4.20 (0.45)	4.67 (0.04)
HD	ℓ_1-SVM	0.84 (0.03)	0.66 (0.09)	10.00 (1.00)	498.38 (1.36)
	CESVM$_1$	0.84 (0.04)	0.67 (0.11)	10.00 (1.00)	498.38 (1.36)
	GFS-CESVM$_1$	0.84 (0.03)	0.66 (0.08)	9.80 (0.45)	266.84 (9.59)
TD	**ℓ_1-SVM**	0.99 (0.01)	0.98 (0.02)	1.00 (0.00)	11.41 (0.00)
	CESVM$_1$	1.00 (0.01)	0.99 (0.01)	1.00 (0.00)	11.41 (0.00)
	GFS-CESVM$_1$	1.00 (0.01)	0.99 (0.01)	1.00 (0.00)	11.41 (0.00)
BCD	ℓ_1-SVM	0.96 (0.01)	0.92 (0.02)	10.40 (0.89)	49.26 (0.28)
	CESVM$_1$	0.95 (0.03)	0.90 (0.05)	2.00 (0.00)	4.18 (0.00)
	GFS-CESVM$_1$	0.95 (0.03)	0.90 (0.05)	2.00 (0.00)	4.18 (0.00)
CD	**ℓ_1-SVM**	0.84 (0.11)	0.71 (0.21)	19.40 (1.34)	99.37 (0.66)
	CESVM$_1$	0.84 (0.10)	0.69 (0.19)	33.00 (3.87)	177.39 (9.69)
	GFS-CESVM$_1$	0.84 (0.10)	0.69 (0.19)	32.00 (4.06)	172.63 (11.70)
LD	ℓ_1-SVM	0.94 (0.06)	0.88 (0.13)	19.20 (1.79)	99.47 (0.24)
	CESVM$_1$	0.94 (0.06)	0.88 (0.13)	20.20 (1.30)	99.36 (0.33)
	GFS-CESVM$_1$	0.94 (0.06)	0.88 (0.13)	9.00 (0.71)	29.04 (0.82)
PD	ℓ_1-SVM	0.91 (0.07)	0.83 (0.13)	30.60 (3.05)	149.48 (0.80)
	CESVM$_1$	0.92 (0.07)	0.84 (0.14)	30.00 (4.00)	133.44 (12.98)
	GFS-CESVM$_1$	0.92 (0.06)	0.84 (0.11)	9.80 (2.39)	14.77 (0.34)
S1	ℓ_1-SVM	0.87 (0.03)	0.74 (0.06)	78.60 (4.16)	449.51 (0.46)
	CESVM$_1$	0.88 (0.03)	0.76 (0.05)	40.00 (5.48)	192.46 (8.67)
	GFS-CESVM$_1$	0.89 (0.02)	0.78 (0.05)	45.80 (1.30)	199.22 (1.12)
S2	**ℓ_1-SVM**	0.96 (0.01)	0.92 (0.01)	38.60 (3.13)	199.83 (0.20)
	CESVM$_1$	0.96 (0.01)	0.92 (0.01)	44.20 (4.09)	188.38 (6.75)
	GFS-CESVM$_1$	0.96 (0.01)	0.92 (0.01)	42.20 (8.53)	193.84 (5.63)
S3	ℓ_1-SVM	0.92 (0.02)	0.84 (0.04)	129.40 (2.61)	699.05 (1.07)
	CESVM$_1$	0.95 (0.00)	0.90 (0.01)	72.60 (2.30)	407.10 (16.28)
	GFS-CESVM$_1$	0.95 (0.01)	0.90 (0.02)	43.60 (4.16)	198.54 (0.82)



(a) *CKD* (b) *TD*

Fig. 4. Performance behaviors of GFS-RCESVM$_1$ under cost uncertainty on *CKD* and *TD*.

Table 3
Numerical results of the BCP and BB algorithms, i.e., the mean value of five folds.

Dataset	Γ	Ω	B	BB									BCP								
				obj	*gap* (%)	*#fea*	*#sam*	*#node*	*tt* (*s*)	*rt* (*s*)	*pt* (*s*)	*st* (*s*)	*obj*	*gap* (%)	*#fea*	*#sam*	*#node*	*tt* (*s*)	*rt* (*s*)	*pt* (*s*)	*st* (*s*)
CKD	0	0	50	8.0	0.0	320.0	24.0	71.4	0.2	0.0	0.0	0.2	8.0	0.0	137.8	21.0	146.4	0.4	0.0	0.0	0.4
	0	0	100	6.4	0.0	320.0	24.0	77.4	0.2	0.0	0.0	0.2	6.4	0.0	133.2	19.8	89.2	0.2	0.0	0.0	0.2
	4	2	50	8.8	0.0	320.0	24.0	27.8	0.8	0.0	0.0	0.8	8.8	0.0	146.8	21.4	128.4	0.0	0.0	0.0	0.0
	4	2	100	7.3	0.0	320.0	24.0	112.8	0.6	0.0	0.0	0.6	7.3	0.0	140.6	20.8	258.0	0.6	0.0	0.0	0.6
HD	0	0	50	134.6	0.0	242.4	13.0	3.6	0.2	0.0	0.0	0.2	134.6	0.0	235.6	13.0	36.2	0.2	0.0	0.0	0.2
	0	0	100	123.0	0.0	242.4	13.0	21.8	0.2	0.0	0.0	0.2	123.0	0.0	233.0	13.0	51.0	0.4	0.0	0.0	0.4
	2	1	50	134.6	0.0	242.4	13.0	3.6	0.4	0.0	0.2	0.2	134.6	0.0	235.6	13.0	36.2	0.4	0.0	0.0	0.4
	2	1	100	134.6	0.0	242.4	13.0	3.6	0.0	0.0	0.0	0.0	134.6	0.0	235.4	13.0	34.4	0.2	0.0	0.0	0.2
TD	0	0	50	17.5	0.0	427.2	21.0	3.6	0.0	0.0	0.0	0.0	17.5	0.0	162.8	19.0	8.0	0.4	0.0	0.0	0.4
	0	0	100	17.4	0.0	427.2	21.0	3.4	0.0	0.0	0.0	0.0	17.4	0.0	163.8	19.6	4.2	0.0	0.0	0.0	0.0
	4	2	50	17.7	0.0	427.2	21.0	9.0	0.0	0.0	0.0	0.0	17.7	0.0	164.2	19.0	23.6	0.2	0.0	0.0	0.2
	4	2	100	17.4	0.0	427.2	21.0	3.4	0.2	0.0	0.0	0.2	17.4	0.0	164.0	19.6	4.4	0.0	0.0	0.0	0.0
BCD	0	0	50	28.2	0.0	455.2	30.0	416.8	2.2	0.0	0.2	2.0	28.2	0.0	179.8	29.4	626.8	1.6	0.0	0.0	1.6
	0	0	100	27.8	0.0	455.2	30.0	27.8	0.2	0.0	0.0	0.2	27.8	0.0	170.6	28.2	31.8	0.0	0.0	0.0	0.0
	6	3	50	29.1	0.0	455.2	30.0	1,265.2	4.0	0.0	0.0	4.0	29.1	0.0	188.0	29.6	1,972.2	5.4	0.0	0.0	5.4
	6	3	100	27.8	0.0	455.2	30.0	46.8	1.0	0.0	0.0	1.0	27.8	0.0	171.0	29.0	111.6	0.4	0.0	0.0	0.4
CD	0	0	150	3.1	0.1	49.6	2,000.0	1,547.2	393.2	0.0	0.4	392.8	3.1	0.0	48.0	176.4	4,816.8	35.2	0.0	0.0	35.2
	0	0	300	3.1	0.0	49.6	2,000.0	1.0	4.8	0.0	0.4	4.4	3.1	0.0	46.8	88.2	1.0	0.0	0.0	0.0	0.0
	400	41	150	3.2	3.4	49.6	2,000.0	1,176.6	600.2	0.0	0.2	600.0	3.2	1.2	48.8	430.4	29,868.8	485.6	0.0	0.0	485.6
	400	41	300	3.1	0.0	49.6	2,000.0	1.0	5.4	0.0	0.4	5.0	3.1	0.0	46.8	88.8	1.0	0.2	0.0	0.0	0.2
LD	0	0	150	1.6	0.5	57.6	7,070.0	90.0	601.0	0.0	1.0	600.0	1.6	0.0	57.4	406.8	42,472.0	600.0	0.0	0.0	600.0
	0	0	300	1.6	0.0	57.6	7,070.0	1.0	62.2	0.0	1.2	61.0	1.6	0.0	57.0	105.6	1.0	0.2	0.0	0.0	0.2
	1414	40	150	1.7	1.2	57.6	7,070.0	69.2	601.0	0.0	1.0	600.0	1.6	0.1	57.0	359.4	34,160.8	520.2	0.0	0.0	520.2
	1414	40	300	1.7	3.9	57.6	7,070.0	41.0	601.4	0.0	1.4	600.0	1.7	0.8	57.6	443.4	28,189.4	600.0	0.0	0.0	600.0
PD	0	0	150	2.7	1.2	81.6	5,966.0	159.4	492.2	0.0	1.8	490.4	2.7	0.4	77.0	438.6	19,790.2	480.0	0.0	0.0	480.0
	0	0	300	2.7	0.0	81.6	5,966.0	1.0	72.0	0.0	2.0	70.0	2.7	0.0	73.4	116.2	1.0	0.6	0.0	0.0	0.6
	1193	39	150	2.9	8.7	81.6	5,966.0	39.6	602.0	0.0	2.0	600.0	2.8	2.3	77.8	534.4	18,502.4	600.0	0.0	0.0	600.0
	1193	39	300	2.7	0.0	81.6	5,966.0	66.2	384.2	0.0	1.6	382.6	2.7	0.0	73.8	152.4	116.8	3.0	0.0	0.0	3.0
S1	0	0	500	33.8	48.8	800.0	2,000.0	101.6	605.4	0.0	5.4	600.0	32.8	43.3	527.4	731.2	1,110.8	600.0	0.0	0.0	600.0
	0	0	1000	22.5	0.0	800.0	2,000.0	85.0	220.0	0.0	5.0	215.0	22.5	0.0	514.6	466.2	107.4	74.8	0.0	0.0	74.8
	400	38	500	74.2	231.3	800.0	2,000.0	58.6	605.4	0.0	5.4	600.0	45.9	101.2	527.2	761.2	827.4	600.0	0.0	0.0	600.0
	400	38	1000	23.9	5.8	800.0	2,000.0	164.8	605.2	0.0	5.2	600.0	23.3	3.5	524.6	708.0	1,623.6	600.0	0.0	0.0	600.0
S2	0	0	500	144.8	0.0	1,600.0	1,000.0	70.8	266.0	0.0	5.2	260.8	144.8	0.0	814.8	340.8	551.6	136.6	0.0	0.0	136.6
	0	0	1000	144.8	0.0	1,600.0	1,000.0	23.0	172.6	0.0	5.0	167.6	144.8	0.0	809.6	313.2	19.0	28.2	0.0	0.0	28.2
	200	36	500	151.4	4.6	1,600.0	1,000.0	102.2	605.4	0.0	5.4	600.0	146.3	1.0	817.2	573.0	2,616.4	600.0	0.0	0.0	600.0
	200	36	1000	144.8	0.0	1,600.0	1,000.0	84.0	327.6	0.0	5.6	322.0	144.8	0.0	809.6	322.0	57.6	69.2	0.0	0.0	69.2
S3	0	0	500	60.4	20.1	1,200.0	1,500.0	99.2	607.0	0.0	7.0	600.0	58.2	15.8	668.4	732.4	1,737.4	600.2	0.0	0.2	600.0
	0	0	1000	50.0	0.0	1,200.0	1,500.0	8.8	140.6	0.0	7.0	133.6	50.0	0.0	658.0	381.2	1.0	23.8	0.0	0.0	23.8
	300	37	500	86.2	73.5	1,200.0	1,500.0	70.0	606.2	0.0	6.2	600.0	72.5	44.8	668.4	720.6	1,729.2	600.0	0.0	0.0	600.0
	300	37	1000	50.2	0.3	1,200.0	1,500.0	235.8	607.0	0.0	7.0	600.0	50.1	0.3	662.0	541.8	2,958.4	520.0	0.0	0.0	520.0

MCC produces an informative and truthful statistical result in evaluating binary classification (Chicco & Jurman, 2020).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{5.1}$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}. \tag{5.2}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively. The number of selected features evaluates how many features are used for prediction, while the total cost measures how much cost is spent to acquire the features. In Table 2, the best model is highlighted in bold regarding the predictive and economic performances on each dataset. The results show that ℓ_1-SVM is inferior to CESVM$_1$ and GFS-CESVM$_1$, and this outcome emphasizes the importance of the feature cost incorporated into models, such that the cost is considered in the model training. CESVM$_1$ provides competitive outcomes in several datasets; however, GFS-CESVM$_1$ reliably attains a high predictive performance at low cost. For further result analysis, visual illustration is shown in Figs. 3, in which the accuracy and the number of selected features are measured under varying budgets.

For low-dimension datasets such as *CKD, HD, TD*, and *BCD*, both CESVM$_1$ and GFS-CESVM$_1$ provide a high predictive performance. It appears that ℓ_1-SVM spends a higher cost to achieve the accuracy comparable to the other models, especially for *CKD* and *BCD*. For high-dimension datasets such as *CD, LD,* and *PD*, the results vary depending on the datasets because their sample size is much smaller than feature dimensions. This could lead to an overfitting issue that a model is trained with features that arise from noise in the data, rather than the underlying distribution, which results in unreliable predictive performances. The accuracy of the models greatly fluctuates, and therefore, no models cannot be said to outperform the other models for *CD, LD,* and *PD*. To rigorously appraise the models, synthetic datasets are tested that have enough features and samples, and the significant outcomes are more clear. The results demonstrate that GFS-CESVM$_1$ consistently achieves higher accuracy and also reaches a plateau with a lower cost than the other models, especially for *S1* and *S3*, while all the models show similar performances for *S2*. With regard to the number of selected features, GFS-CESVM$_1$ acquires more features at the same cost than the other models due to a discounted group price. This is observed over the datasets but clearly appeared in *HD, BCD*, and *PD*. Overall, these results highlight the importance of considering the feature cost and the group structure of features to obtain better predictive and economic performances.

5.3. Experimental results of robustness to cost uncertainty

This section investigates the influence of cost uncertainty on the classification performance through GFS-RCESVM$_1$ that takes into account robustness to the cost uncertainty of individual and grouped features. GFS-RCESVM$_1$ identifies optimal and feasible solutions that are more immune to the cost uncertainty and prevent the predictive performance from being significantly deteriorated. A few examples are tested with the different cost uncertainty levels of Γ and Ω, specifically for 0%, 10%, and 20% of the original feature and group sizes. When $\Gamma = 0$ and $\Omega = 0$, GFS-RCESVM$_1$ is identical to GFS-CESVM$_1$. Fig. 4 depicts classification accuracy and the number of selected features under the different uncertainty scenarios in *CKD* and *TD*. Although solutions become more conservative as Γ and Ω increase, GFS-RCESVM$_1$ attempts to maintain its predictability with the best subset of features under the uncertainty. For instance, the accuracy of 0.94 is reached at the cost of 3 when no cost uncertainty exists in *CKD*, while a similar predictive performance is attained at the cost of 5 when the cost uncertainty exists. Similar behavior is observed in *TD*. Therefore, GFS-RCESVM$_1$ enables practitioners to make a desirable budget plan to achieve high classification accuracy when the cost variation exists.

5.4. Experimental results of proposed branch-Cut-and-Price algorithm

This section evaluates the proposed BCP algorithm in solving GFS-RCESVM$_1$ on various benchmark and synthetic datasets. To validate its significance, the BCP algorithm is also compared to the ordinary BB algorithm with different settings of Γ, Ω, and B. Two different scenarios of cost uncertainty (Γ, Ω) and budget (B) are conducted on the experimental datasets because they influence computational complexity. Γ and Ω are tested with 0% and 20% of the original feature and group sizes, while B varies across datasets because the number of important features and their costs are different. The problems become more difficult to solve as Γ and Ω increase and B decreases; higher Γ and Ω mean more cost variation, and lower B represents a tighter budget. Six criteria are evaluated, which include the objective value (obj), integrality gap (gap), the number of features ($\#fea$), samples ($\#sam$), and nodes ($\#node$) considered in a search tree, and total time (tt). The obj indicates the quality of a solution, and gap shows the closeness of a solution to global optimality. The $\#fea$ and $\#sam$ represent model complexity ($com = \#row + \#col$), while $\#node$ shows how many nodes are explored which indicates model efficiency. Lastly, tt measures how long the algorithm takes to solve a problem and represents a computational expense.

Table 3 shows the numerical result of the BB and BCP algorithms. Also, Table 4 summarizes how much improvement can be achieved through the BCP algorithm from the BB algorithm; i.e., $\Delta(\%) = \frac{\text{BB}-\text{BCP}}{\text{BB}} \times 100$. All the evaluation criteria prefer a smaller value except $\#node$; that is, the negative Δ for $\#node$ and the positive Δ for the remaining measures indicate that the proposed BCP algorithm attains a better performance than the BB algorithm. The results are highlighted in bold in Table 4 where the BCP algorithm outperforms the BB algorithm. We also decompose tt into reading time (rt), pre-solving time (pt), and solving time (st). To statistically validate the result of experiments conducted with various datasets, the pairwise t-test is subsequently performed with 0.05 of a significance level (α). Table 5 summarizes t-test results with t-statistic (t) and p-value (p) where the statistical significance of the observed difference is highlighted in bold, i.e., $p \leq 0.05$.

Low-dimension datasets, such as $CKD, HD, TD,$ and BCD, have considerably lower computational complexity than the other datasets, and both BCP and BB algorithms solve the problems within 1 s. As a result, obj and gap in the algorithms are zero. Although the BCP algorithm reduces model complexity (i.e., $\#fea$ and $\#sam$) and explores more $\#node$, it does not statistically improve tt as compared to the BB algorithm. It suggests that the BCP algorithm is not advantageous for small-scale problems. To investigate the validity of BCP further, high-dimension datasets are tested, which include $CD, LD,$ and PD. Due to a large number of features in high-dimension datasets, the BB and BCP algorithms cannot solve several problems within the time limit for problem-solving, which is 600 s. The results show that the BCP algorithms dramatically reduce $\#fea$ and therefore process more $\#node$. In many cases, the BCP algorithm can solve problems quicker than the BB algorithm and also achieve better obj and gap when both algorithms do not reach solution optimality. Specifically, the BCP algorithm, on average, achieves 0.9% better obj, 40.8% lower gap, and 60.3% shorter tt by lowering 93.7% of $\#fea$ for the high-dimension datasets. Due to the significant imbalance between the feature dimensions and sample size in prior datasets, synthetic datasets are tested. The results show that the BCP algorithm is notably beneficial to achieve improved obj and gap within shorter tt by reducing $\#sam$ and $\#fea$ and processing more $\#node$. The BCP algorithm, on average,

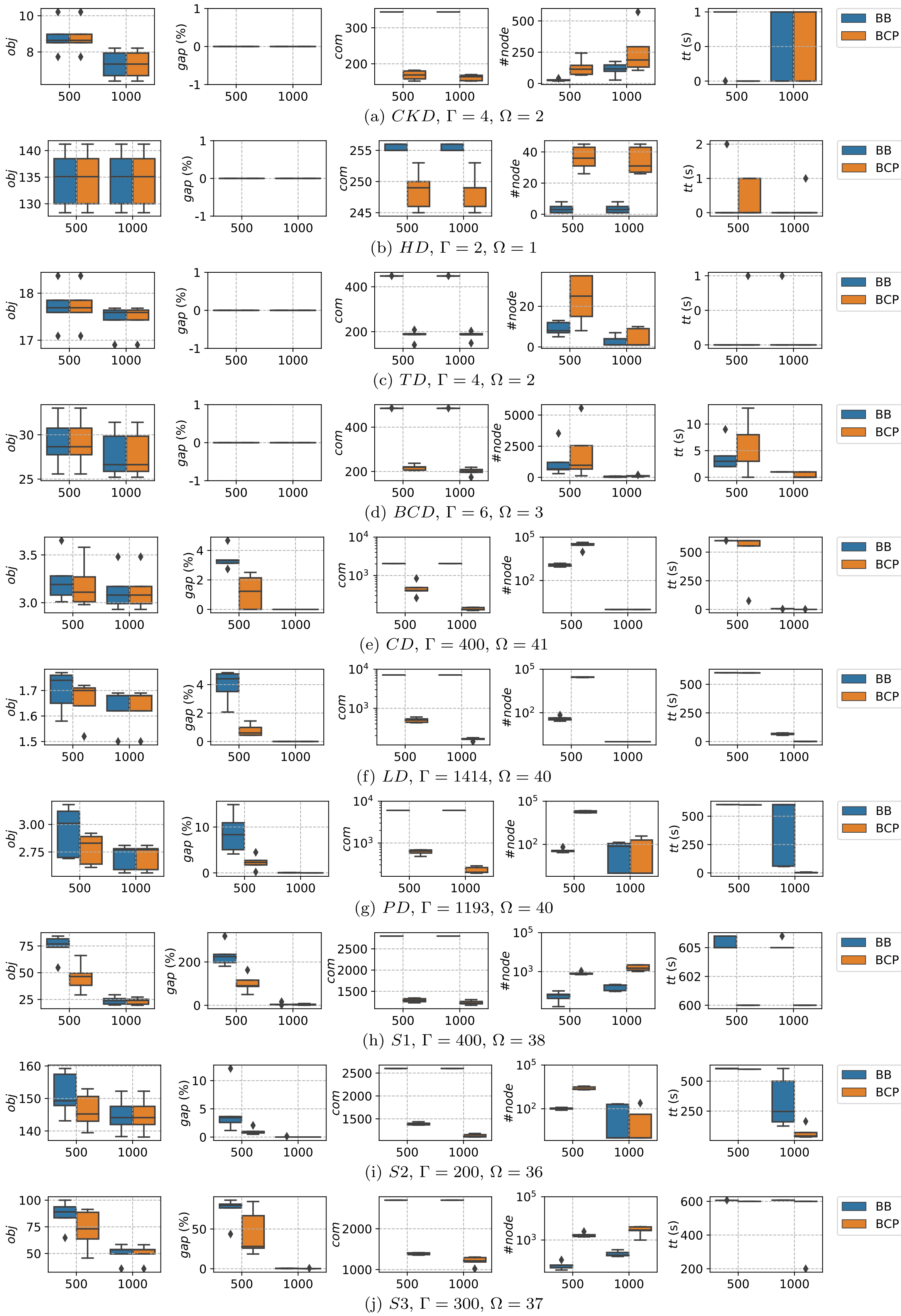


Fig. 5. Performance comparison of BB and BCP algorithms.

Table 4
Performance improvement of the BCP algorithm from the BB algorithm.

Dataset	Γ	Ω	*B*	△ Difference = $\frac{BB-BCP}{BB} \times 100$ (%)								
				obj	*gap*	*#row*	*#col*	*#node*	*tt*	*rt*	*pt*	*st*
CKD	0	0	50	0.0	0.0	**56.9**	**12.5**	**-180.9**	-20.0	0.0	0.0	-20.0
	0	0	100	0.0	0.0	**58.4**	**17.5**	**-114.0**	0.0	0.0	0.0	0.0
	4	2	50	0.0	0.0	**54.1**	**10.8**	**-394.2**	**80.0**	0.0	0.0	**80.0**
	4	2	100	0.0	0.0	**56.1**	**13.3**	**-169.8**	0.0	0.0	0.0	0.0
HD	0	0	50	0.0	0.0	**2.8**	0.0	**-1,794.0**	0.0	0.0	0.0	0.0
	0	0	100	0.0	0.0	**3.9**	0.0	**-188.2**	-20.0	0.0	0.0	-20.0
	2	1	50	0.0	0.0	**2.8**	0.0	**-1,794.0**	-20.0	0.0	20.0	-20.0
	2	1	100	0.0	0.0	**2.9**	0.0	**-1,771.5**	-20.0	0.0	0.0	-20.0
TD	0	0	50	0.0	0.0	**61.9**	**9.5**	**-55.7**	-40.0	0.0	0.0	-40.0
	0	0	100	0.0	0.0	**61.7**	**6.7**	**-17.9**	0.0	0.0	0.0	0.0
	4	2	50	0.0	0.0	**61.6**	**9.5**	**-189.2**	-20.0	0.0	0.0	-20.0
	4	2	100	0.0	0.0	**61.6**	**6.7**	**-20.7**	**20.0**	0.0	0.0	**20.0**
BCD	0	0	50	0.0	0.0	**60.5**	**2.0**	**-62.0**	**31.7**	0.0	**20.0**	**25.0**
	0	0	100	0.0	0.0	**62.5**	**6.0**	**-166.2**	**20.0**	0.0	0.0	20.0
	6	3	50	0.0	0.0	**58.7**	**1.3**	**-64.2**	-28.9	0.0	0.0	-28.9
	6	3	100	0.0	0.0	**62.4**	**3.3**	**-515.0**	**60.0**	0.0	0.0	**60.0**
CD	0	0	150	0.0	**60.0**	**3.2**	**91.2**	**-155.0**	**89.9**	0.0	**40.0**	**88.9**
	0	0	300	0.0	0.0	**5.7**	**95.6**	0.0	**100.0**	0.0	**40.0**	**100.0**
	400	41	150	**1.6**	**66.7**	**1.6**	**78.5**	**-2,664.0**	**19.1**	0.0	**20.0**	**19.1**
	400	41	300	0.0	0.0	**5.7**	**95.6**	0.0	**96.7**	0.0	**40.0**	**96.7**
LD	0	0	150	**0.2**	100.0	**0.3**	**94.2**	**-53,186.2**	**0.2**	0.0	**100.0**	0.0
	0	0	300	0.0	0.0	**1.0**	**98.5**	0.0	**99.7**	0.0	**100.0**	**99.7**
	1414	40	150	**2.5**	**78.0**	0.0	**93.7**	**-76,033.5**	**0.2**	0.0	**100.0**	0.0
	1414	40	300	0.0	0.0	**1.0**	**98.5**	0.0	**99.0**	0.0	**100.0**	**99.0**
PD	0	0	150	**0.6**	**53.8**	**5.6**	**92.6**	**-17,120.7**	**20.2**	0.0	**100.0**	**20.0**
	0	0	300	0.0	0.0	**10.0**	**98.1**	0.0	**99.2**	0.0	**100.0**	**99.2**
	1193	39	150	**5.3**	**70.9**	**4.6**	**91.0**	**-50,322.5**	**0.3**	0.0	**100.0**	0.0
	1193	39	300	0.0	**60.0**	**9.5**	**97.4**	**-48.8**	**99.2**	0.0	**100.0**	**99.2**
S1	0	0	500	**2.2**	-3.1	**34.1**	**63.4**	**-1,558.9**	**0.9**	0.0	**100.0**	0.0
	0	0	1000	0.0	0.0	**35.7**	**76.7**	4.0	**68.0**	0.0	**100.0**	**67.3**
	400	38	500	**38.6**	**55.6**	**34.1**	**61.9**	**-1,952.0**	**0.9**	0.0	**100.0**	0.0
	400	38	1000	**2.0**	**35.1**	**34.4**	**64.6**	**-1,020.4**	**0.9**	0.0	**100.0**	0.0
S2	0	0	500	0.0	**20.0**	**49.1**	**65.9**	**-154.3**	**68.0**	0.0	**100.0**	**67.5**
	0	0	1000	0.0	0.0	**49.4**	**68.7**	6.9	**81.3**	0.0	**100.0**	**80.4**
	200	36	500	**3.3**	**48.7**	**48.9**	**42.7**	**-2,458.8**	**0.9**	0.0	**100.0**	0.0
	200	36	1000	0.0	**20.0**	**49.4**	**67.8**	11.9	**77.2**	0.0	**100.0**	**76.6**
S3	0	0	500	**3.6**	**21.4**	**44.3**	**51.2**	**-1,752.3**	**1.1**	0.0	**97.1**	0.0
	0	0	1000	0.0	0.0	**45.2**	**74.6**	19.5	**83.1**	0.0	**100.0**	**82.2**
	300	37	500	**16.2**	**38.3**	**44.3**	**52.0**	**-2,537.0**	**1.0**	0.0	**100.0**	0.0
	300	37	1000	**0.1**	**28.0**	**44.8**	**63.9**	**-1,252.2**	**14.3**	0.0	**100.0**	**13.3**

Table 5
Pairwise *t*-test results between BB and BCP algorithms.

Dataset	*obj*		*gap*		*#fea*		*#sam*		*#node*		*tt*	
	t	*p*	*t*	*p*	*t*	*p*	*t*	*p*	*t*	*p*	*t*	*p*
CKD	-	-	-	-	**78.1**	**0.00**	**10.3**	**0.00**	**-3.3**	**0.00**	0.9	0.38
HD	-	-	-	-	**6.0**	**0.00**	-	-	**-14.6**	**0.00**	-0.6	0.58
TD	-	-	-	-	**60.2**	**0.00**	**6.2**	**0.00**	**-2.7**	**0.02**	-1.0	0.33
BCD	-	-	-	-	**81.9**	**0.00**	**3.9**	**0.00**	-2.0	0.06	0.0	1.00
CD	**2.2**	**0.04**	**2.3**	**0.03**	**4.7**	**0.00**	**45.5**	**0.00**	**-2.6**	**0.02**	**2.5**	**0.02**
LD	**2.6**	**0.02**	**2.8**	**0.01**	**3.2**	**0.00**	**174.5**	**0.00**	**-4.2**	**0.00**	**4.5**	**0.00**
PD	**2.3**	**0.04**	**2.4**	**0.03**	**7.6**	**0.00**	**120.0**	**0.00**	**-3.8**	**0.00**	**2.5**	**0.02**
S1	**2.4**	**0.03**	**2.3**	**0.03**	**76.1**	**0.00**	**46.2**	**0.00**	**-6.0**	**0.00**	**2.8**	**0.01**
S2	1.5	0.14	1.5	0.15	**305.1**	**0.00**	**23.2**	**0.00**	**-2.7**	**0.01**	**4.2**	**0.00**
S3	2.0	0.06	2.0	0.06	**69.7**	**0.00**	**25.6**	**0.00**	**-5.7**	**0.00**	**2.5**	**0.02**

achieves 5.5% better *obj*, 22.0% lower *gap*, and 33.1% shorter *tt* by lowering 42.8% of *#fea* and 62.8% of *#sam* for the synthetic datasets.

The BB and BCP algorithms are intuitively compared in terms of their performances under 20% of cost uncertainty in Fig. 5. The result is aligned with what was discussed earlier. Overall, the experimental results demonstrate that the BCP algorithm can reduce model complexity, and therefore, it dominantly outperforms the ordinary BB algorithm for finding better solutions a short period of time in difficult problems.

6. Conclusion and future work

In this research, we propose GFS-CESVM$_1$ and GFS-RCESVM$_1$ by considering the group structure of features. Although the integrity in the model captures the discrete nature of decisions (i.e., feature and group selection) while guaranteeing optimal solutions, which cannot be determined by LP, model complexity is escalated with enormous features and constraints. This research proposes the BCP algorithm that considers only a significantly small number of features and constraints in a solution search tree to essentially tackle the complexity issue. To demonstrate the effectiveness of the proposed MILP SVM models and BCP algorithm, the experiments are conducted on various medical diagnosis benchmark and synthetic datasets. The parameter setting of Γ, Ω, and B is varied to test different scenarios.

Firstly, GFS-CESVM$_1$ is compared with ℓ_1-SVM and CESVM$_1$ to show the significance of the feature cost and group feature selection consideration. The results show that ℓ_1-SVM is inferior to CESVM$_1$ and GFS-CESVM$_1$ in many datasets, and this outcome emphasizes the importance of the feature cost incorporated into the model. CESVM$_1$ provides competitive outcomes in some datasets; however, GFS-CESVM$_1$ can reliably attain a high predictive performance at a low cost. Afterward, GFS-RCESVM$_1$ is used to investigate the influence of cost uncertainty on the classification performance. It is noticed that a solution becomes more conservative as the cost uncertainty level increases; thus, GFS-RCESVM$_1$ facilitates practitioners to make a desirable budget plan to achieve high classification accuracy under cost uncertainty. Lastly, to validate the effectiveness of the BCP algorithm, it is compared to the standard BB algorithm in terms of six evaluation criteria: *obj*, *gap*, #*fea*, #*sam*, #*node* and *time*. Through the experimental study, the results show that the proposed BCP algorithm can efficiently solve GFS-RCESVM$_1$ in most of the datasets, especially high-dimension and synthetic datasets, by considering significantly reduced features and/or constraints. Therefore, the BCP algorithm dominantly outperforms the ordinary BB algorithm for finding better objective value and integrality gap and also quickly solve difficult problems.

This research focuses mainly on healthcare applications; however, our proposed models can be applied to other domains such as image analysis and text mining, where numerous features are generally presented. Also, more constraints for sequential feature selection can be incorporated into the model to make it more realistic. The robust and cost-effective feature selection concept can also be applied to other machine learning or deep learning models such as logistic regression, Naive Bayes classifier, random forest, and deep neural network.

References

Achterberg, T., Koch, T., & Martin, A. (2005). Branching rules revisited. *Operations Research Letters, 33*(1), 42–54.

Asuncion, A., & Newman, D. (2007). UCI machine learning repository. http://www.ics.uci.edu/~mlearn/MLRepository.html.

Belotti, P., Lee, J., Liberti, L., Margot, F., & Wächter, A. (2009). Branching and bounds tighteningtechniques for non-convex MINLP. *Optimization Methods & Software, 24*(4–5), 597–634.

Ben-Tal, A., & Nemirovski, A. (2000). Robust solutions of linear programming problems contaminated with uncertain data. *Mathematical Programming, 88*(3), 411–424.

Bennett, K. P., Demiriz, A., & Shawe-Taylor, J. (2000). A column generation algorithm for boosting. In *Icml* (pp. 65–72). Citeseer.

Bertsimas, D., & Sim, M. (2004). The price of robustness. *Operations Research, 52*(1), 35–53.

Bi, J., Zhang, T., & Bennett, K. P. (2004). Column-generation boosting methods for mixture of kernels. In *Proceedings of the Tenth ACM SIGKDD International Conference On Knowledge Discovery and Data Mining* (pp. 521–526).

Bolón-Canedo, V., Porto-Díaz, I., Sánchez-Maroño, N., & Alonso-Betanzos, A. (2014). A framework for cost-based feature selection. *Pattern Recognition, 47*(7), 2481–2489.

Bradley, P. S., & Mangasarian, O. L. (2000). Massive data discrimination via linear support vector machines. *Optimization Methods and Software, 13*(1), 1–10.

Carrizosa, E., Martin, B., & Morales, D. (2006). A column generation approach for support vector machines. *Technical Report*. Technical report.

Chen, J., Zhou, S., Kang, Z., & Wen, Q. (2020). Locality-constrained group lasso coding for microvessel image classification. *Pattern Recognition Letters, 130*, 132–138.

Chicco, D., & Jurman, G. (2020). The advantages of the matthews correlation coefficient (MCC) over f1 score and accuracy in binary classification evaluation. *BMC Genomics, 21*(1), 1–13.

Crone, S. F., Lessmann, S., & Stahlbock, R. (2006). The impact of preprocessing on data mining: An evaluation of classifier sensitivity in direct marketing. *European Journal of Operational Research, 173*(3), 781–800.

Ding, H., Feng, P.-M., Chen, W., & Lin, H. (2014). Identification of bacteriophage virion proteins by the ANOVA feature selection and analysis. *Molecular Biosystems, 10*(8), 2229–2235.

Du, C., Du, C., Zhe, S., Luo, A., He, Q., & Long, G. (2016). Bayesian group feature selection for support vector learning machines. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining* (pp. 239–252). Springer.

Elssied, N. O. F., Ibrahim, O., & Osman, A. H. (2014). A novel feature selection based on one-way ANOVA f-test for e-mail spam classification. *Research Journal of Applied Sciences, Engineering and Technology, 7*(3), 625–638.

Freitas, A., Costa-Pereira, A., & Brazdil, P. (2007). Cost-sensitive decision trees applied to medical data. In *International Conference on Data Warehousing and Knowledge Discovery* (pp. 303–312). Springer.

Friedman, J., Hastie, T., & Tibshirani, R.. A note on the group lasso and a sparse group lasso. arXiv preprint arXiv:1001.0736.

Gamrath, G., Fischer, T., Gally, T., Gleixner, A. M., Hendel, G., Koch, T., Maher, S. J., Miltenberger, M., Müller, B., Pfetsch, M. E. et al. (2016). The scip optimization suite 3.2,.

Hernández-Lobato, D., Hernández-Lobato, J. M., & Dupont, P. (2013). Generalized spike-and-slab priors for bayesian group feature selection using expectation propagation. *The Journal of Machine Learning Research, 14*(1), 1891–1945.

Huo, Y., Xin, L., Kang, C., Wang, M., Ma, Q., & Yu, B. (2020). Sgl-svm: A novel method for tumor classification via support vector machine with sparse group lasso. *Journal of Theoretical Biology, 486*, 110098.

Kumar, M., Rath, N. K., Swain, A., & Rath, S. K. (2015). Feature selection and classification of microarray data using mapreduce based ANOVA and k-nearest neighbor. *Procedia Computer Science, 54*, 301–310.

Lee, I. G., Zhang, Q., Yoon, S. W., & Won, D. (2020). A mixed integer linear programming support vector machine for cost-effective feature selection. *Knowledge-Based Systems*, 106145.

Ling, C. X., Yang, Q., Wang, J., & Zhang, S. (2004). Decision trees with minimal costs. In *Proceedings of the 21st international conference on machine learning* (p. 69). ACM.

Liu, J., Lin, Y., Wu, S., & Wang, C. (2018). Online multi-label group feature selection. *Knowledge-Based Systems, 143*, 42–57.

Maldonado, S., Pérez, J., & Bravo, C. (2017). Cost-based feature selection for support vector machines: An application in credit scoring. *European Journal of Operational Research, 261*(2), 656–665.

Maldonado, S., Pérez, J., Weber, R., & Labbé, M. (2014). Feature selection for support vector machines via mixed integer linear programming. *Information Sciences, 279*, 163–175.

Meier, L., Van De Geer, S., & Bühlmann, P. (2008). The group lasso for logistic regression. *Journal of the Royal Statistical Society: Series B (Statistical Methodology), 70*(1), 53–71.

Piramuthu, S. (2004). Evaluating feature selection methods for learning in data mining applications. *European Journal of Operational Research, 156*(2), 483–494.

Simon, N., Friedman, J., Hastie, T., & Tibshirani, R. (2013). A sparse-group lasso. *Journal of Computational and Graphical Statistics, 22*(2), 231–245.

Tang, F., Adam, L., & Si, B. (2018). Group feature selection with multiclass support vector machine. *Neurocomputing, 317*, 42–49.

Turney, P. (2000). Types of cost in inductive concept learning. in: workshop on cost-sensitive learning. In *17th international conference on machine learning university* (pp. 15–25).

Turney, P. D. (1994). Cost-sensitive classification: Empirical evaluation of a hybrid genetic decision tree induction algorithm. *Journal of Artificial Intelligence Research, 2*, 369–409.

Unler, A., & Murat, A. (2010). A discrete particle swarm optimization method for feature selection in binary classification problems. *European Journal of Operational Research, 206*(3), 528–539.

Wang, J., Wang, M., Li, P., Liu, L., Zhao, Z., Hu, X., & Wu, X. (2015). Online feature selection with group structure analysis. *IEEE Transactions on Knowledge and Data Engineering, 27*(11), 3029–3041.

Yang, J., & Honavar, V. (1998). Feature subset selection using a genetic algorithm. In *Feature extraction, construction and selection* (pp. 117–136). Springer.

Yuan, M., & Lin, Y. (2006). Model selection and estimation in regression with grouped variables. *Journal of the Royal Statistical Society: Series B (Statistical Methodology), 68*(1), 49–67.

Zhang, L., & Zhou, W. (2013). Analysis of programming properties and the row–column generation method for 1-norm support vector machines. *Neural Networks, 48*, 32–43.

Zhou, Q., Zhou, H., & Li, T. (2016). Cost-sensitive feature selection using random forest: Selecting low-cost subsets of informative features. *Knowledge-Based Systems, 95*, 1–11.

Zhu, J., Rosset, S., Tibshirani, R., & Hastie, T. J. (2004). 1-norm support vector machines. In *Advances in Neural Information Processing Systems* (pp. 49–56).